

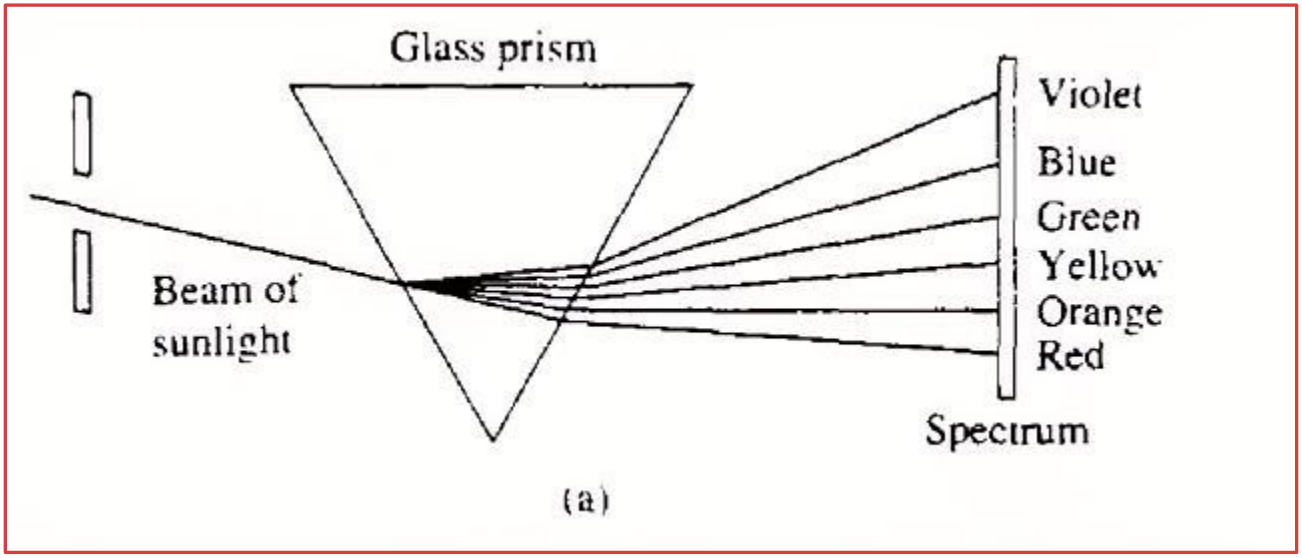
ICE-3107: Digital Signal Processing

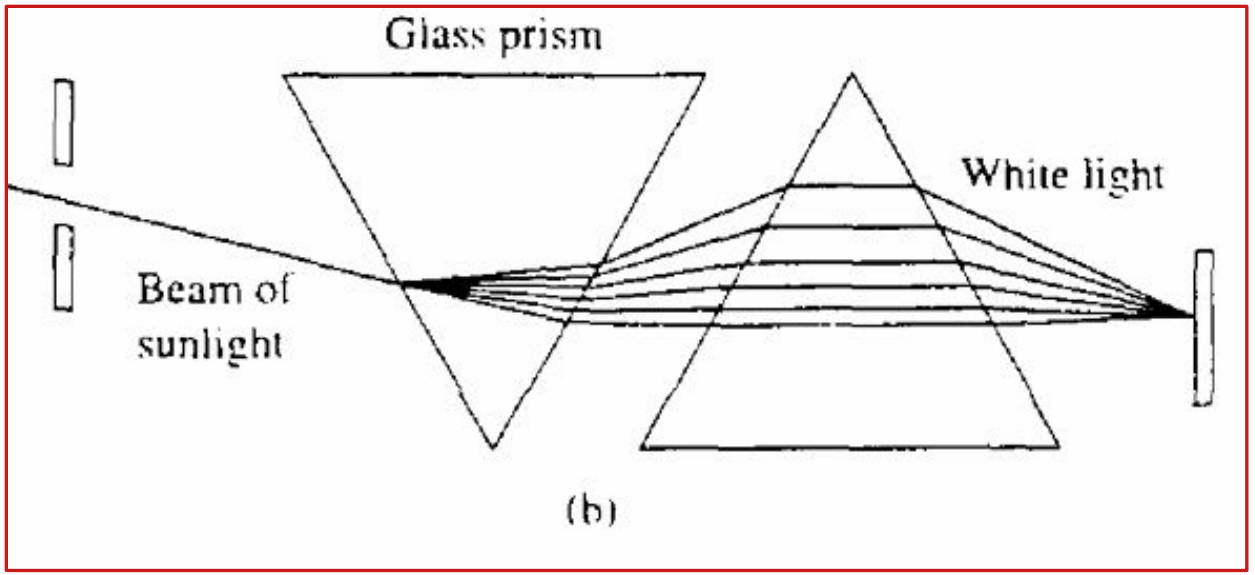
Fourier Series and Fourier Transform

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Lecture Outcomes

- **Understand** the concept, properties, and applications of the Fourier Series, Fourier Transform, and DFT.


Glass prism
Violet
Blue
Green
Yellow
Orange
Red
Beam of
sunlight
Spectrum
(a)


Glass prism
White light
Beam of
sunlight
(b)

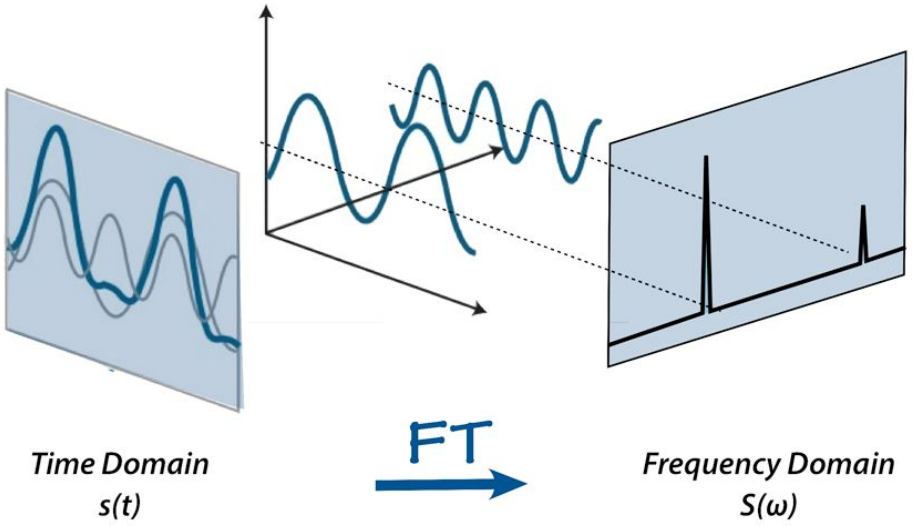
FREQUENCY ANALYSIS
OF
CONTINUOUS-TIME SIGNALS

Introduction

Why do we need Fourier analysis?

In signal processing, many signals look complicated in time domain but become much simpler when we study them in terms of sinusoids or complex exponentials.

The central idea is: Any "reasonable" signal can be built from sine and cosine waves of different frequencies.



Introduction (Cont.)

So, Fourier methods help us answer:

- What frequencies are present in a signal?
- How strong is each frequency?
- How does a system change each frequency?
- How can we reconstruct the original signal from frequency components?

Basic mathematical tools

✓ Periodic signal

A signal $x(t)$is **periodic** if there exists a positive number T such that

$$x(t+T) = x(t) \; for \; all \; t$$

The smallest such T is called the **fundamental period**.

The corresponding **fundamental frequency** is

$$f_0 = \frac{1}{T}$$

and the **fundamental angular frequency** is

$$\omega_0 = \frac{2\pi}{T}$$

Meaning

- T: time taken for one full repetition.
- f_0 :cycles per second.
- ω_0 :radians per second.

Basic mathematical tools (Cont.)

✓ Orthogonality idea

Fourier analysis works because different sinusoids are **orthogonal** over one period.

This means their inner product becomes zero.

For example, over one period T,

$$\int_{t_0}^{t_0+T} \cos(m\omega_0 t)\cos(n\omega_0 t)\, dt = 0 \; if \; m \neq n$$

Similarly for sine and cosine mixed terms.

This is very important because orthogonality allows us to "extract" each frequency component independently.

Basic mathematical tools (Cont.)

✓ **Euler's formula**

A key identity is:

$$e^{j\theta} = \cos\theta + j\sin\theta$$

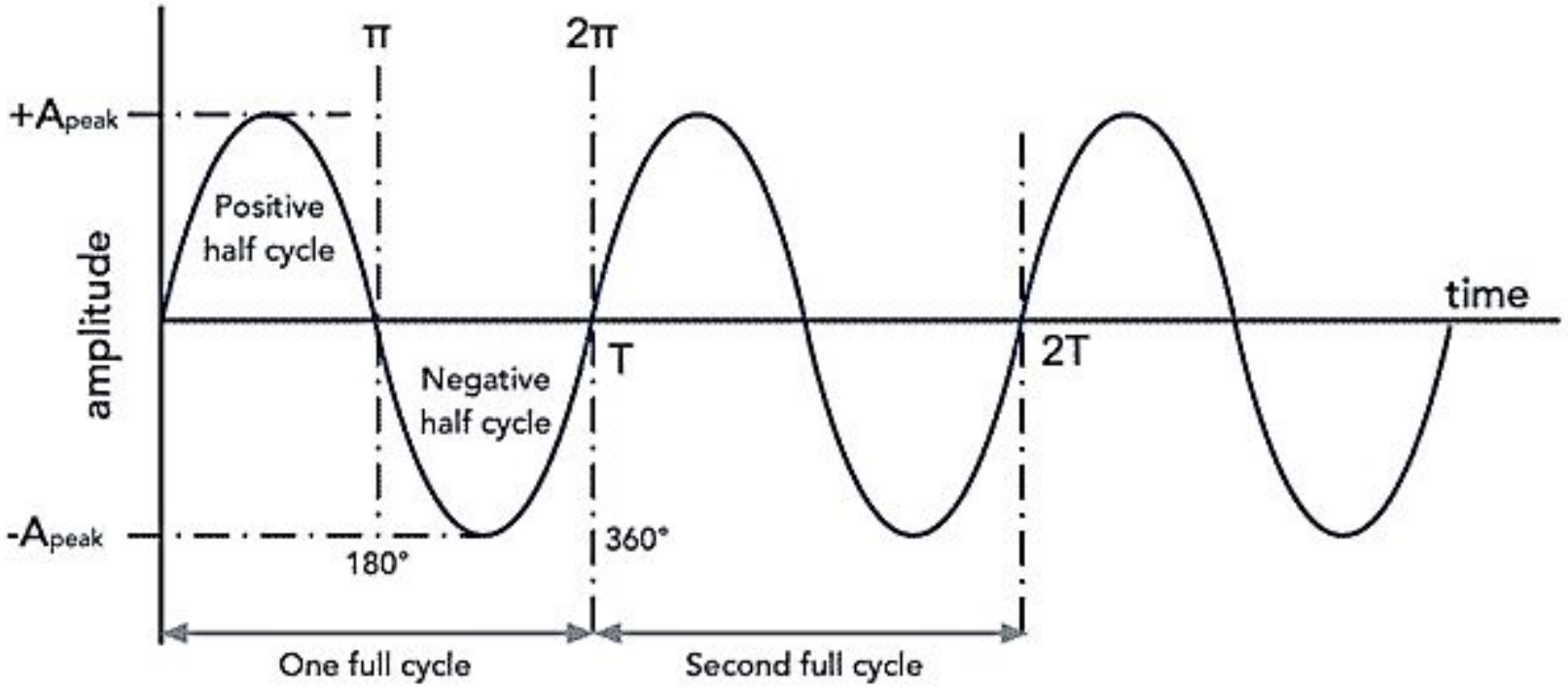
So,

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Fourier Series

- Fourier series applies to periodic signals.
- It says a periodic signal can be represented as a sum of harmonically related sinusoids.



Trigonometric Fourier Series

A periodic signal $x(t)$ with period T can be written as

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$$

where:

$$\omega_0 = \frac{2\pi}{T}$$

The coefficients are

$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} x(t)\, dt$$

$$a_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \cos(n\omega_0 t)\, dt$$

$$b_n = \frac{2}{T} \int_{t_0}^{t_0+T} x(t) \sin(n\omega_0 t)\, dt$$

Trigonometric Fourier Series (Cont.)

Meaning of coefficients

- a_0 average or DC value of the signal
- a_n :amount of cosine at frequency $n\omega_0$
- b_n :amount of sine at frequency $n\omega_0$

The frequencies in the expansion are not arbitrary. They are multiples of the fundamental frequency:

$$\omega_0,\ 2\omega_0,\ 3\omega_0,\ \ldots$$

These are called **harmonics**.

Practice Problem 01

A signal is represented by

$$x(t) = 2\cos(\omega_0 t) + \frac{1}{2}\cos(3\omega_0 t) + \frac{1}{4}\cos(5\omega_0 t)$$

Answer:

1. Which harmonic is strongest?
2. Which harmonic contributes most to waveform sharpness?
3. What happens if all harmonics except the first are removed?

Practice Problem 01

1. Compare the amplitudes:

- $2\cos(\omega_0 t) \rightarrow$ amplitude 2
- $\frac{1}{2}\cos(3\omega_0 t) \rightarrow$ amplitude $\frac{1}{2}$
- $\frac{1}{4}\cos(5\omega_0 t) \rightarrow$ amplitude $\frac{1}{4}$

So, the strongest harmonic is the **1st harmonic.**

2. Higher harmonics create faster oscillations and sharper changes in the waveform.

1st harmonic: smoothest

3rd harmonic: sharper

5th harmonic: sharpest

Therefore, the 5th harmonic contributes most to sharpness.

Practice Problem 01

3. The waveform becomes a pure cosine wave. It becomes smoother and less complex.

Complex Exponential Fourier Series

Fourier Series is used to represent periodic signals as a sum of complex exponentials.

Mathematical Representation:

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$$

Where:

- $\omega_0 = \frac{2\pi}{T}$ →fundamental angular frequency
- C_k →Fourier coefficients

Complex Exponential Fourier Series (Cont.)

Fourier Coefficient

$$C_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt$$

Meaning

- C_k tells how much of the complex exponential at frequency $k\omega_0$ exists in the signal.
- k can be positive or negative.
- Positive and negative frequency terms are both needed in complex form.

Parseval's relation

Energy over one period can be expressed in terms of Fourier coefficients:

$$\frac{1}{T}\int_{t_0}^{t_0+T} |x(t)|^2\, dt = \sum_{k=-\infty}^{\infty} |C_k|^2$$

Meaning

The total average power of a periodic signal is equal to the sum of the powers of all harmonic components.

This is one of the most useful results in Fourier theory.

Parseval's relation: Practice Problem

- Given

$$x(t) = A\cos(\omega_0 t)$$

verify Parseval's relation.

- Also verify by

$$x(t) = A\sin(\omega_0 t)$$

Symmetry properties in Fourier series

❖ Even signals

If $x(t)$ is even:

$$x(t) = x(-t)$$

then all sine coefficients are zero:

$$b_n = 0$$

So, the Fourier series contains only cosine terms.

❖ Odd signals

If $x(t)$ is odd:

$$x(t) = -x(-t)$$

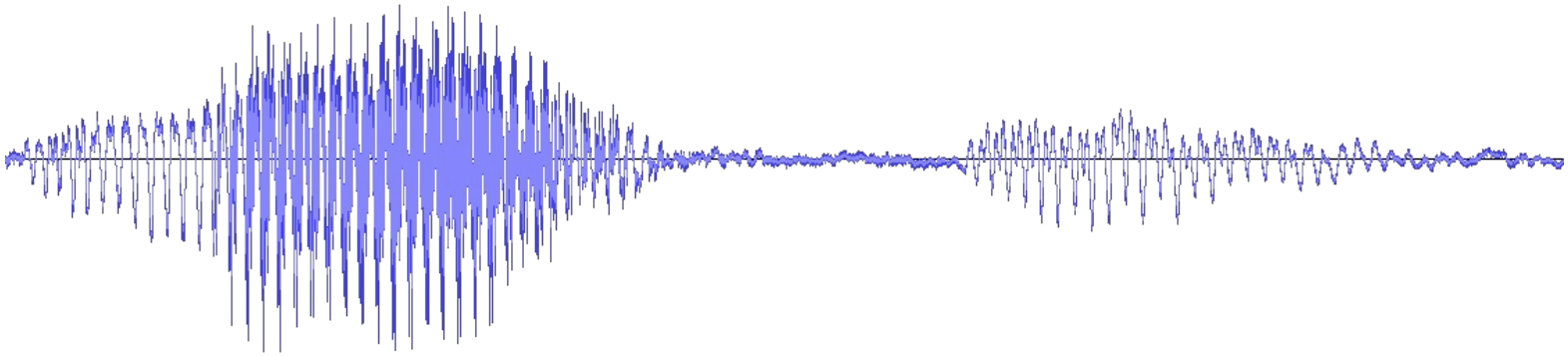
then all cosine coefficients and DC term are zero:

$$a_0 = 0, a_n = 0$$

So, the Fourier series contains only sine terms.

Fourier Transform

Fourier series is for periodic signals.
Fourier transform is for aperiodic signals.
It gives a full frequency representation of signals that do not repeat.



Fourier Transform (Cont.)

For a signal $x(t)$,the Fourier transform is

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt$$

The inverse Fourier transform is

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t}\, d\omega$$

Meaning

- $x(t)$ is the time-domain signal.
- $X(\omega)$ is the frequency-domain representation.
- $X(\omega)$ tells how much of each angular frequency exists in the signal.

Unlike Fourier series, which gives discrete frequency lines, Fourier transform gives a ***continuous spectrum****.*

Properties of Fourier Transform

- Linearity Property

If

$$x(t) \leftrightarrow X(\omega)$$

&

$$y(t) \leftrightarrow Y(\omega)$$

Then linearity property states that

$$ax(t)+by(t) \leftrightarrow aX(\omega)+bY(\omega)$$

Properties of Fourier Transform (Cont.)

- Time Shifting Property

If

$$x(t) \leftrightarrow X(\omega)$$

Then Time shifting property states that

$$x(t-t0) \leftrightarrow e^{-j\omega t0} X(\omega)$$

Properties of Fourier Transform (Cont.)

- Time Reversal Property

If

$$x(t) \leftrightarrow X(\omega)$$

Then Time reversal property states that

$$x(-t) \leftrightarrow X(-\omega)$$

Properties of Fourier Transform (Cont.)

❖ Time Scaling Property

If

$$x(t) \leftrightarrow X(\omega)$$

Then Time scaling property states that

$$x(at) = \frac{1}{|a|} X\left(\frac{w}{a}\right)$$

Existence conditions

A sufficient condition for Fourier Transform existence is:

$$\int_{-\infty}^{\infty} |x(t)| \, dt < \infty$$

This means the signal must be absolutely integrable.

Important Fourier Transform pairs

- Find the Fourier Transform of

$$x(t) = \delta(t)$$

Ans: $X(\omega)=e^{-j\omega\cdot 0}=1$

- Find the Fourier Transform of

$$x(t) = e^{-at}u(t), a > 0$$

where $u(t)$is the unit step.

Ans: $X(\omega)=\frac{1}{a+jw}$

Digital systems process samples, not continuous-time signals.

So, for discrete signals we need:

1. **DTFT** for discrete-time infinite sequences
2. **DFT** for finite-length sequences

Main Difference

DTFT	DFT
• Used for infinite-length sequences	• Used for finite-length sequences
• Frequency is continuous	• Frequency is discrete
• Output is continuous curve	• Output is finite points
• Mostly theoretical analysis	• Mostly practical/computer implementation
• Not directly computable on computers	• Easily computable using FFT

Discrete Time Fourier Transforms (DTFT)

The discrete-time Fourier transform (DTFT) or the Fourier transform of a discrete-time sequence x[n] is a representation of the sequence in terms of the complex exponential sequence $e^{j\omega n}$

For a discrete-time sequence $x[n]$,

$$X(w) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

DTFT (Cont.)

Here, $X(\omega)$ is a complex function of real frequency variable ω and it can be written as

$$X(\omega)=X_{re}(\omega)+jX_{img}(\omega)$$

Where $X_{re}(\omega)$, $X_{img}(\omega)$ are real and imaginary parts of $X(\omega)$ respectively.

$$X_{re}(\omega)=|X(\omega)|\cos\theta(\omega)$$

$$X_{img}(\omega)=|X(\omega)|\sin\theta(\omega)$$

$$|X(\omega)|^2=|X_{re}(\omega)|^2+|X_{im}(\omega)|^2$$

Discrete Fourier Transform (DFT)

For a finite sequence $x[n]$,$n = 0,1, \dots, N-1$,the DFT is

$$X[k] = \sum_{n=0}^{N-1} x\,[n] e^{-j\frac{2\pi}{N}kn} \quad k = 0,1, \dots, N-1$$

The inverse DFT is

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X\,[k] e^{j\frac{2\pi}{N}kn} \quad n = 0,1, \dots, N-1$$

Discrete Fourier Transform (DFT)

Problem: Find the DFT of {1,1,1,1}
Ans: {4,0,0,0}

Problem: Find the DFT of (1,-1,1,-1}
Ans: {0,0,4,0}

5.2.2 Multiplication of Two DFTs and Circular Convolution

Suppose that we have two finite-duration sequences of length N, $x_1(n)$and $x_2(n)$.

Their respective N-point DFTs are

$$X_1(k) = \sum_{n=0}^{N-1} x_1(n) e^{-j2\pi nk/N}$$

$$X_2(k) = \sum_{n=0}^{N-1} x_2(n) e^{-j2\pi nk/N}$$

Circular Convolution:

$$x_3(m) = \sum_{n=0}^{N-1} x_1(n) x_2\left(\left((m-n)\right)_N\right)$$

Example 5.2.1

- Perform the circular convolution of the following two sequences:

$$x_1(n) = \{2, 1, 2, 1\}$$
$$x_2(n) = \{1, 2, 3, 4\}$$

Ans: y(n)={14, 16, 14, 16}

Fast Fourier Transform

The DFT equation is:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

For every value of k, we need many:

- multiplications,
- additions.

If N is large, the computation becomes very slow.

The Fast Fourier Transform (FFT) is one of the most important algorithms in Digital Signal Processing (DSP).

FFT is a fast computational method used to calculate the Discrete Fourier Transform (DFT).

Fast Fourier Transform

The DFT is:

$$X[k] = \sum_{n=0}^{N-1} x\,[n] W_N^{kn}$$

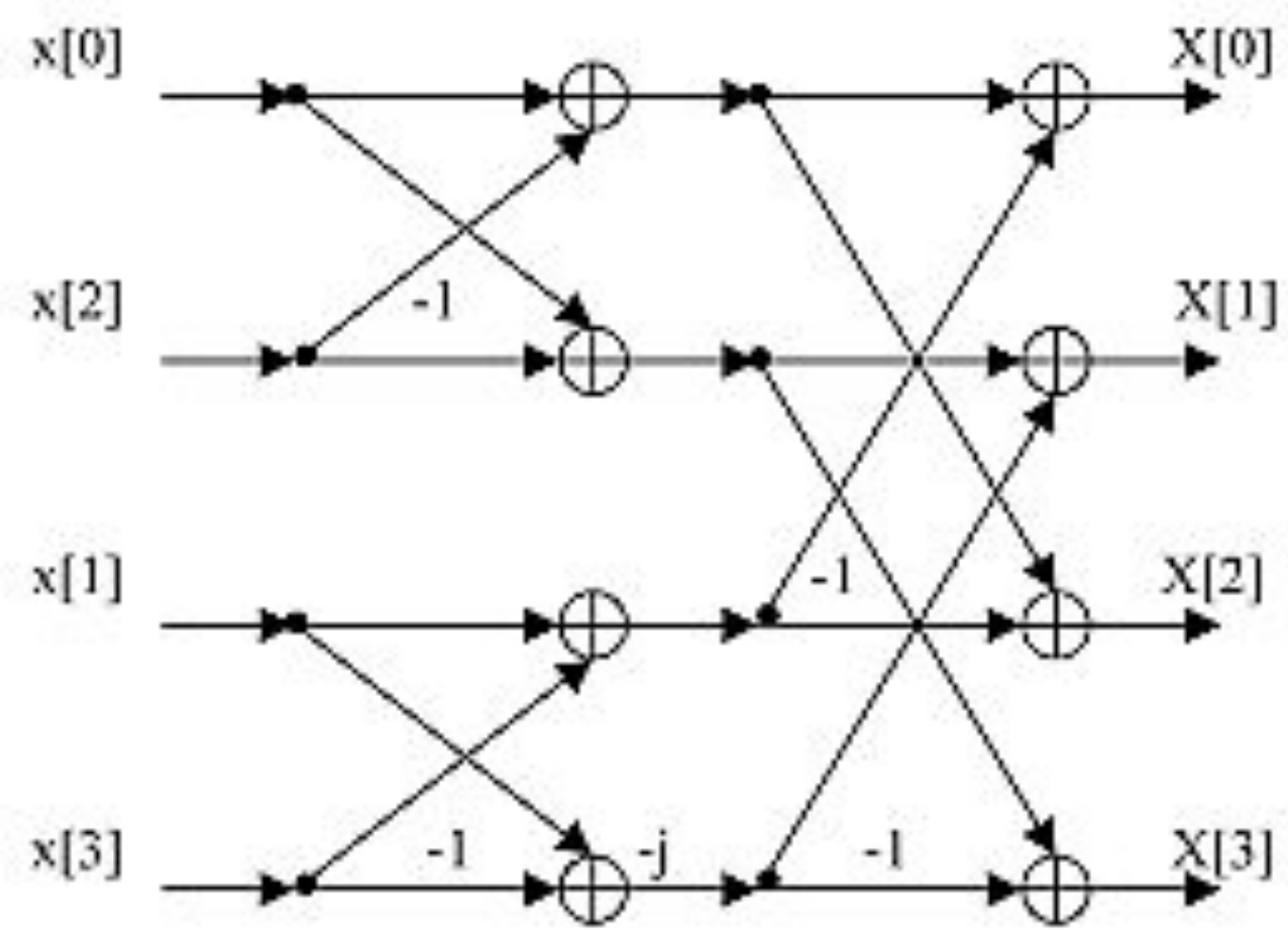
where:

$$W_N = e^{-j2\pi/N}$$

called the **twiddle factor**.

Decimation In Time (DIT-FFT)

4 Point DIT



DIT-FFT (Cont.)

Problem: Find the DFT of x(n)={1, 2, 3, 4} using 4-Point DIT FFT.

Ans: X(k)={10, -2+2j, -2, -2-2j}

Goertzel Algorithm

- The Goertzel Algorithm is a DSP algorithm used to compute:
  - one specific DFT frequency component,
  - or a few selected frequency components,
    without calculating the entire DFT.

- It is extremely useful when:
  - only a small number of frequencies are needed,
  - full FFT computation is unnecessary.

Goertzel Algorithm (Cont.)

$$X[k] = s[N-1] - e^{-j\frac{2\pi k}{N}} s[N-2]$$

s[n] = internal state

Initial conditions:

$$s[-1] = 0, s[-2] = 0$$

Goertzel Algorithm Steps

Find:

$$\omega_k = \frac{2\pi k}{N}$$

Compute:

$$a_k = 2\cos(\omega_k)$$

Initialize:

$$s[-1] = 0, s[-2] = 0$$

Use recursion:

$$s[n] = x[n] + a_k s[n-1] - s[n-2]$$

Compute:

$$X[k] = s[N-1] - e^{-j\omega_k} s[N-2]$$

Practice Problem

Problem: Find the DFT coefficient $X[1]$of:

$$x[n] = \{1, 2, 3, 4\}$$

using Goertzel algorithm.

Ans: 2+2j

Thank You


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